

Jeremy G. Baier | Curriculum Vitae

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Education

PhD	Sept. 2022 - Present
<i>Oregon State University</i>	<i>Corvallis, Oregon</i>
MS Physics	Sept. 2022 - Mar. 2025
<i>Oregon State University</i>	<i>Corvallis, Oregon</i>
BA Mathematics and Physics	May 2022
<i>Kenyon College</i>	<i>Gambier, Ohio</i>

Funding & Awards

Oregon Lottery Graduate Scholarship	2024
(\$3,000)	
NASA Oregon Space Grant Consortium Graduate Fellowship	2023
(\$10,000)	
APS DGRAV Travel Award	2022
(\$300)	
Kenyon Summer Sciences Grant	2021
(\$4,000)	
Kenyon Summer Sciences Grant	2020
(retracted due to Covid-19)	

Observing Proposals

Co-I: "The North American Nanohertz Observatory for Gravitational Waves"	January 2024
◦ Greenbank Telescope, GBT/24B-427	
◦ Status: awarded 2205.0 hours	
Co-I: "The North American Nanohertz Observatory for Gravitational Waves"	January 2024
◦ Very Large Array, VLA/24B-429	
◦ Status: awarded 168.0 hours	
Co-I: "Anomalous Mode Changing in the High-Precision Millisecond Pulsar J1909-3744"	January 2025
◦ Greenbank Telescope, GBT/25B-206	
◦ Status: awarded 2.0 hours	
Co-I: "No Pulsar Left Behind: Confirmation of Student-Discovered Pulsar Candidates"	July 2024
◦ Greenbank Telescope, GBT/25A-369	
◦ Status: awarded 20.0 hours	

Teaching & Mentoring

Teaching Assistant Positions.....	
◦ Graduate Teaching Assistant , Oregon State University,	Fall 2022 - Present
<i>Descriptive Astronomy, General Physics III, General Physics with Calculus</i>	
◦ Teaching Assistant , Kenyon College,	Spring 2020 - Spring 2022
<i>General Physics I, General Physics II, Fields and Spacetime</i>	

Graded student work and maintained gradebook.

Curriculum Development.....

- **PH104 Descriptive Astronomy**, Oregon State University, Spring 2023
 Rewrote the laboratory curriculum to include more hands-on activities
 and incorporate simulated experiments with Universe Sandbox.
 Added a new laboratory on general relativity and gravitational waves.

Undergraduate Student Research Mentoring.....

- *Sammuel Grommes*, Oregon State University 2026
"Advanced noise models for 2 pulsars in the NANOGrav 20 year data set"
- *Caleb Fowler*, Oregon State University 2026
"Exploratory Data Analysis for the NANOGrav 20 year data set"
- *Stephanie Poole*, Oregon State University 2026
"Pulsar Timing for the 3rd IPTA Data Release"
- *Stephanie Poole*, Oregon State University 2025
"Searching for a turnover in the NANOGrav 20 year data set"
- *Kyle Gourlie*, Oregon State University 2024
"Variance in PTA sensitivity curves"
- *Rodney Downer*, Oregon State University 2023
"LISA multi-messenger pipeline"

NANOStars Mentoring Program.....

- Teach undergraduate students the basics of pulsar timing and pulsar candidate ranking. *September 2023-Present*

Leadership & Professional Service

Research leadership.....

- NANOGrav 20-year dataset Noise Modeling Lead *January 2026-Present*
- NANOGrav Timing/Detection Working Group Liaison *May 2025-Present*

Reviewer for international journals.....

- Astronomy & Astrophysics
- MNRAS

Committees.....

- Instructor Hiring Committee at Oregon State *Fall 2024*

Conference organization.....

- Committee Member, Scientific Organizing Committee, IPTA Meeting, Caltech, Pasadena, CA
 June 2025,

Software Development

Lead Developer.....

hasasia, pint_pal

Contributing Developer.....

discovery, ENTERPRISE, PINT, enterprise_extensions, la_forge, Bilby, LALsuite

Invited talks

2. **Gravitational Wave Astronomy North West Meeting**, August 10, 2026
"In their detection era: An overview of Pulsar Timing Arrays"
1. **IPTA-wide Telecon**, December 3, 2025
"Noise Mitigation in NANOGrav"